

Project Report: IoT Vehicle Tracking System

1. Introduction

The objective of this project is to develop an IoT-based vehicle tracking system capable of collecting, transmitting, storing, and visualizing GPS location data in real time. The system monitors geographic coordinates, altitude, speed, heading, and device status, allowing users to track vehicle movements and review historical routes through a web dashboard.

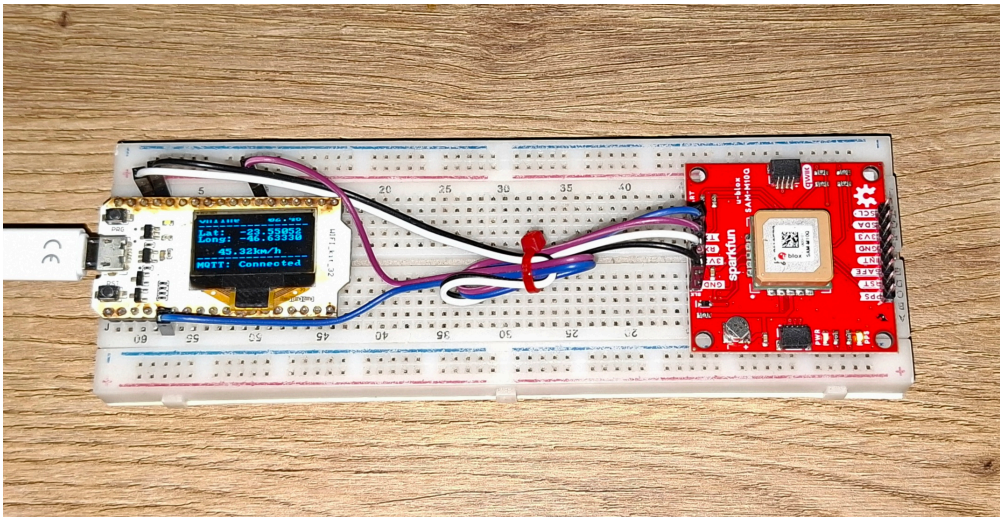


Figure 1 - Hardware Circuit

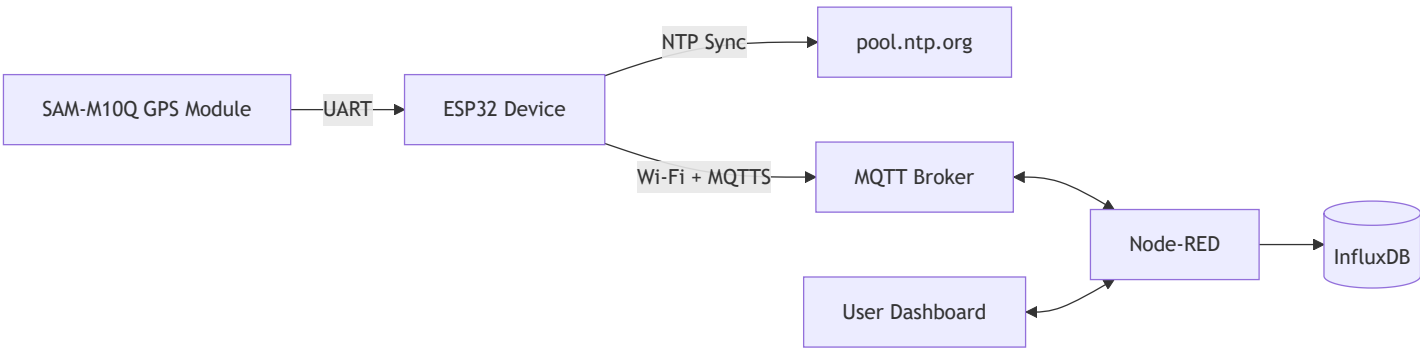
2. System Architecture

The solution is composed of embedded hardware, cloud communication services, a data storage layer, and a visualization interface.

Technologies Used

Component	Description
ESP32	Provides Wi-Fi connectivity and system control
SAM-M10Q GPS Module	Acquires GPS positioning data
OLED Display (128×64)	Displays device status and diagnostic information
ESP-IDF	Development framework for ESP32
FreeRTOS	Real-time operating system used by ESP-IDF
HiveMQ	MQTT broker hosting platform
Node-RED	Message processing, automation, and user dashboard
InfluxDB	Time-series database used to store tracking data

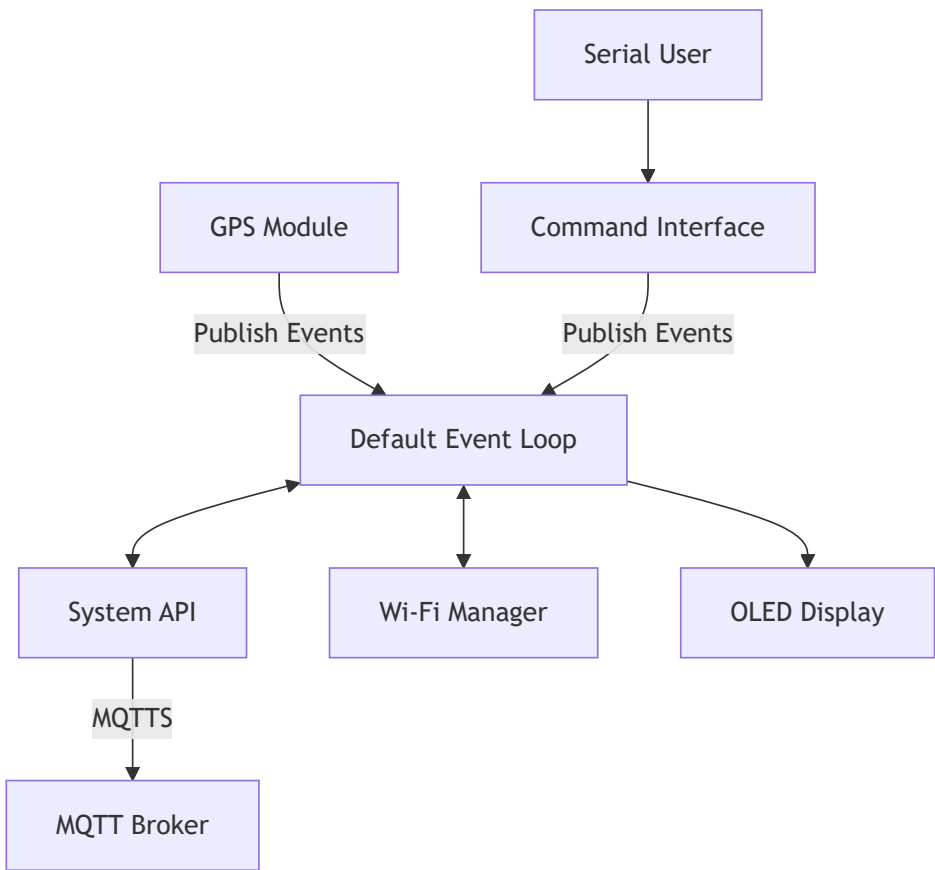
System Diagram



3. ESP32 Application Architecture

The embedded application follows an event-driven architecture using the ESP-IDF Event Loop.

Application Components



Component Responsibilities

GPS Module

Responsible for communication with the SAM-M10Q GPS receiver and generation of location events.

System API

Handles MQTT communication, system synchronization, and internal time management.

Wi-Fi Manager

Responsible for network connectivity and reconnection procedures.

OLED Display

Displays operational information and diagnostics.

Command Interface

Receives commands through the serial interface, such as Wi-Fi credential updates.

4. Security

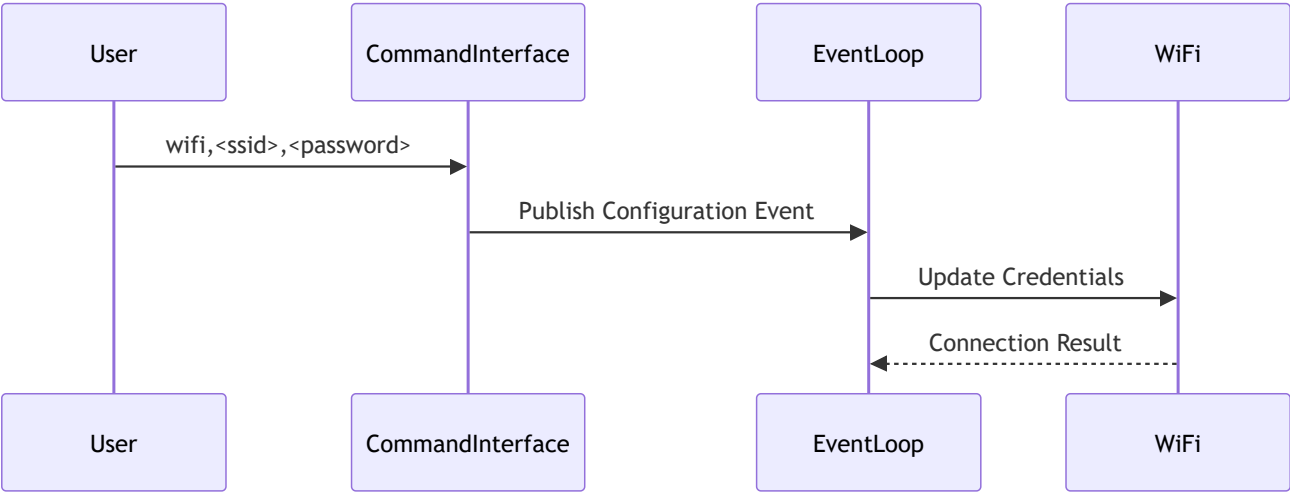
Communication between devices and the MQTT broker uses MQTTS (MQTT over TLS). Certificate validation is performed using the ESP-IDF built-in certificate bundle. MQTT authentication requires valid user credentials.

Node-RED also connects securely to the broker using TLS authentication.

InfluxDB operates locally and is not exposed to external networks, reducing the attack surface of the system.

5. Sequence Diagrams

Wi-Fi Configuration



MQTT Connection and Device Status

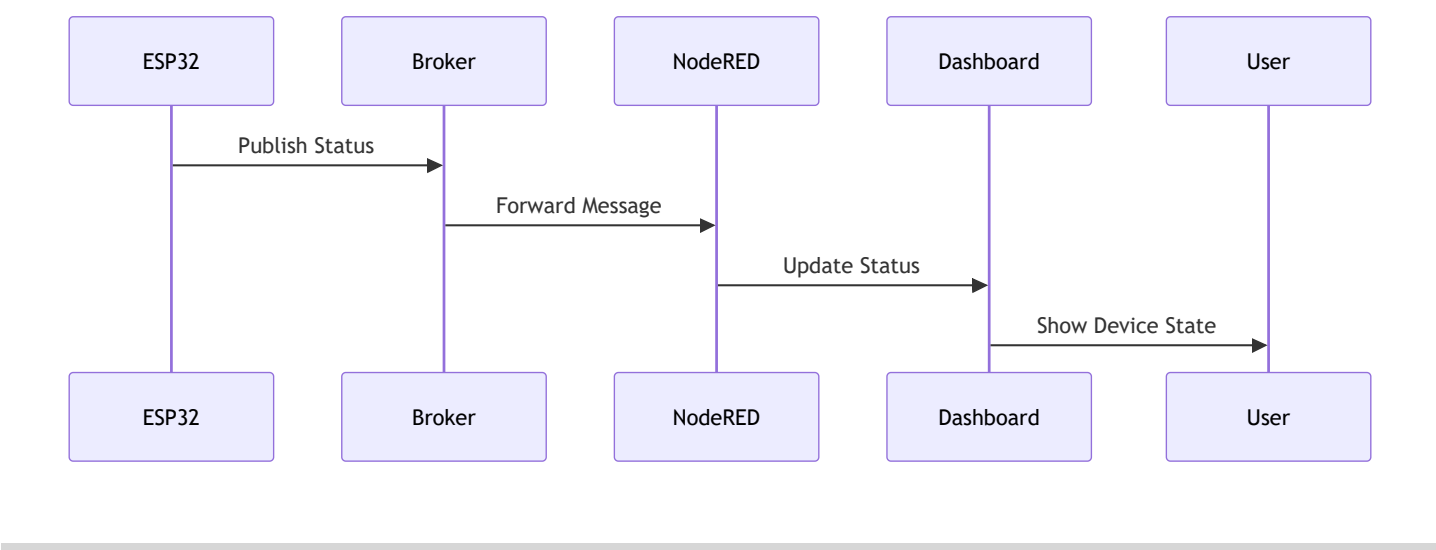
The device publishes its online status using MQTT Last Will and Testament (LWT).

Topic

/tracking_device/<id>/status

Payload

```
{
  "online": true
}
```



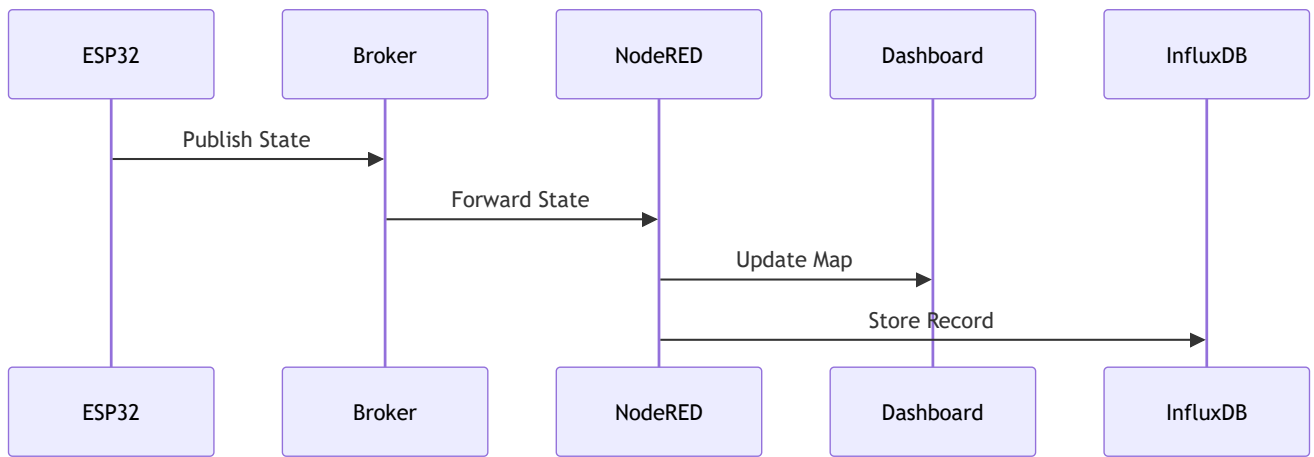
Device State Transmission

Topic

```
/tracking_device/<id>/state
```

Payload

```
{
  "latitude": 0.0,
  "longitude": 0.0,
  "altitude": 0.0,
  "speed_kmh": 0.0,
  "course_deg": 0.0,
  "satellites": 0,
  "hdop": 0,
  "timestamp": 0,
  "time_on": 0
}
```



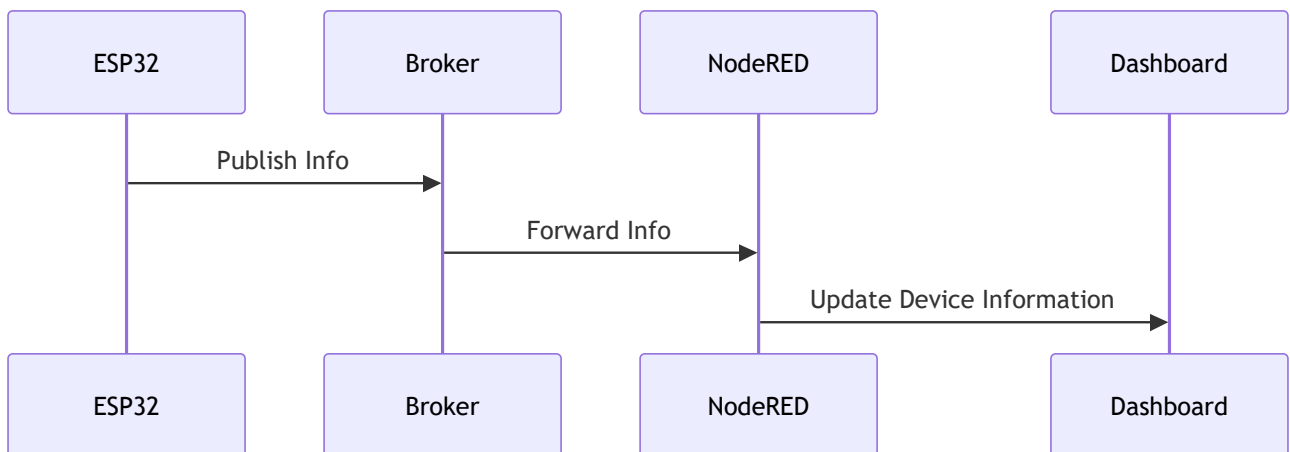
Device Information Transmission

Topic

```
/tracking_device/<id>/info
```

Payload

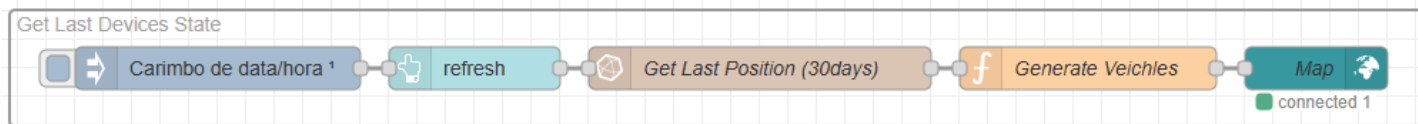
```
{
  "ip": "192.168.1.100",
  "timestamp": 0,
  "time_on": 0
}
```



6. Node-RED Flows

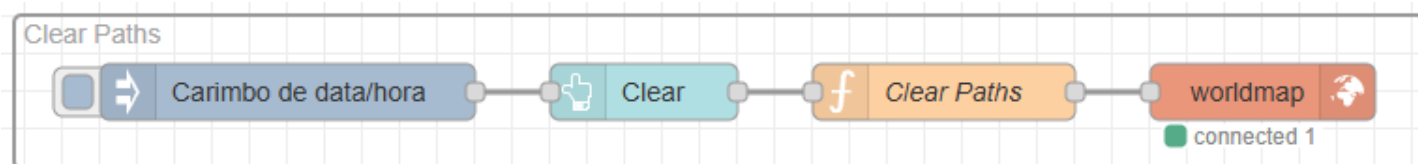
Get Last Device State

Queries InfluxDB for the latest known position of each device within the last 30 days and displays the results on the map.



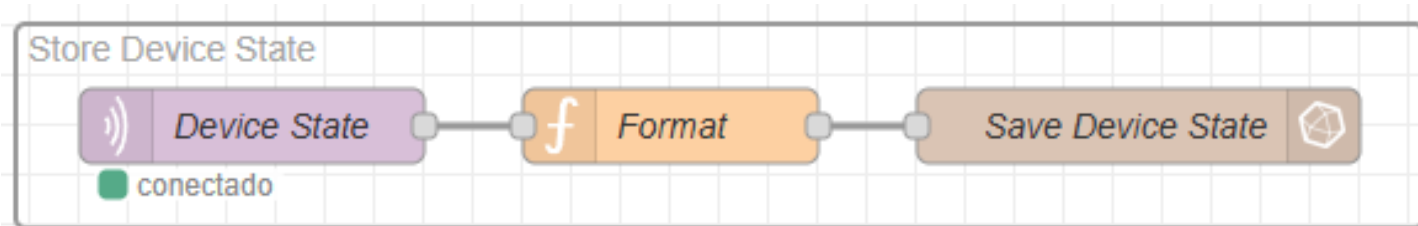
Clear Paths

Removes all displayed routes from the map interface.



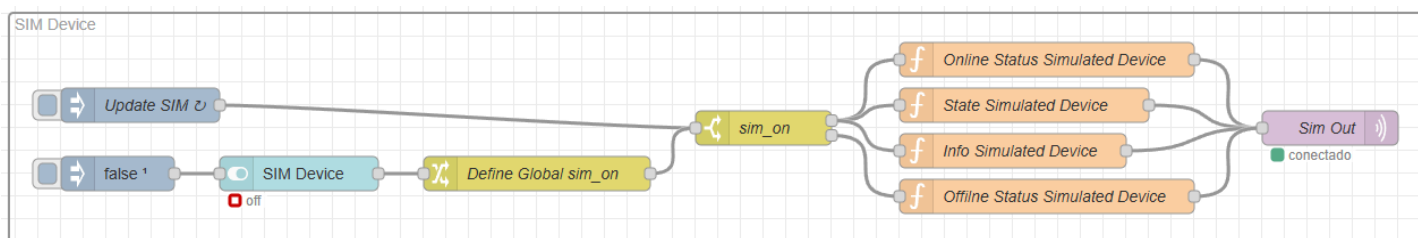
Store Device State

Processes incoming state messages and stores them in InfluxDB.



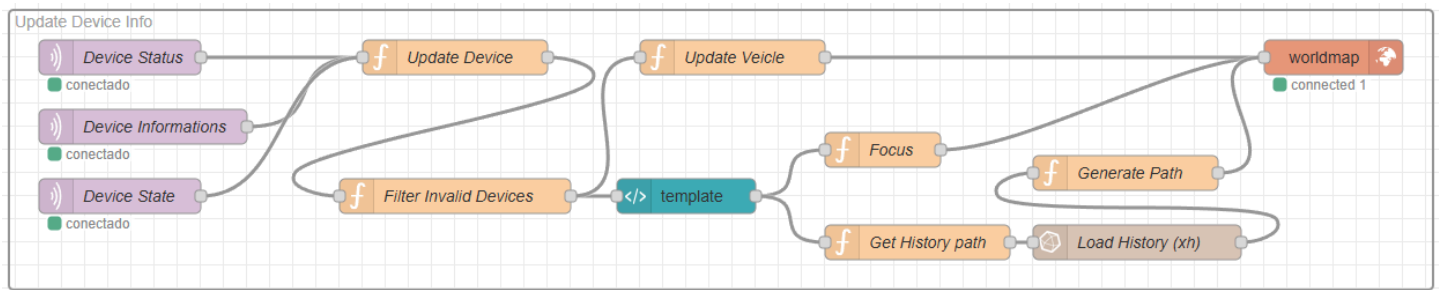
SIM Device

Generates simulated GPS data for testing and demonstration purposes.



Update Device Information

Updates the dashboard with current device information and provides actions such as focusing the map on a selected device or retrieving historical routes.



7. User Interface

No Connected Devices

Dashboard state when no devices are online.

The screenshot shows a dashboard with two main sections: 'Map' and 'Devices'.

Map Section: Features a map of a city street grid. A red car icon is positioned on 'Rua S de Ourem'. Above the map are two buttons: 'CLEAR' and 'REFRESH'. The map includes a zoom control on the left and a street view pegman on the right. At the bottom of the map, it says 'Leaflet | Map data © OpenStreetMap contributors'.

Devices Section: Displays information for a single device named 'sim'. The status is 'OFFLINE'. The IP address is '192.168.1.100'. The location is 'Lat: 41.80625 Lon: -6.75845'. The speed is '45 km/h', the battery level is '12', and the signal strength is '2s'. Below this information is a '1 hours' time range selector and a 'Get History' button.

At the bottom right of the dashboard, there is a 'SIM DEVICE' label and a toggle switch that is currently turned off.

Simulated Device Online

Dashboard displaying an active simulated device.

Map

CLEAR

REFRESH

Leaflet | Map data © OpenStreetMap contributors

Devices

sim

ONLINE

IP

192.168.1.100

Lat: 41.80656

Lon: -6.75834

45 km/h

12

3s

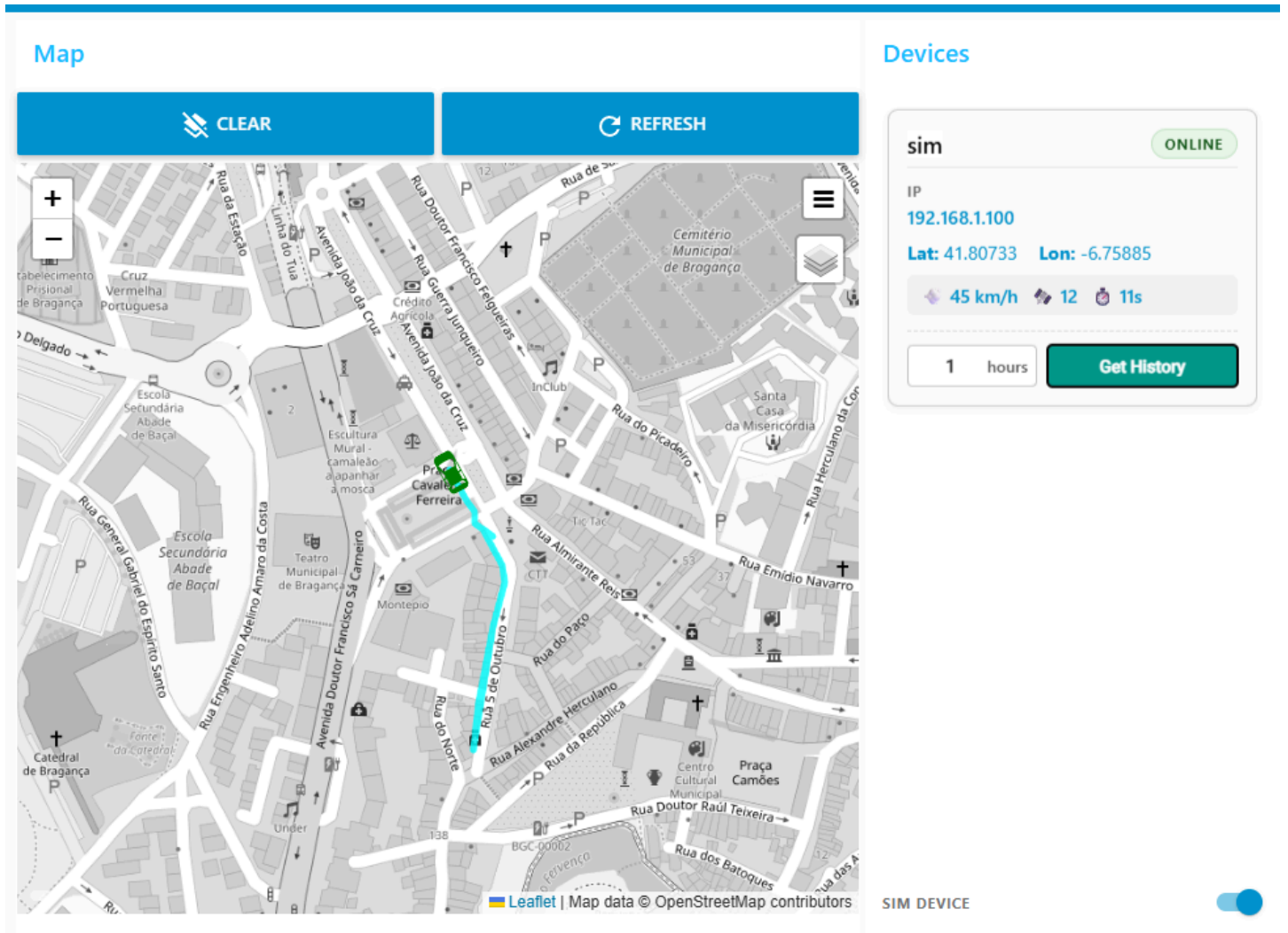
1 hours

Get History

SIM DEVICE

Historical Route Visualization

Dashboard displaying the historical route of the simulated device.



8. Conclusion

The developed system successfully demonstrates an end-to-end IoT tracking solution capable of collecting GPS information, transmitting data securely through MQTT, storing historical records in a time-series database, and providing real-time visualization through a web dashboard. The modular architecture based on ESP-IDF, FreeRTOS, Node-RED, and InfluxDB allows the platform to be easily extended for additional telemetry data, fleet monitoring, and advanced analytics.